

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper 4-space indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into folders such as templates, static, models, datasets, and app.py.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like hdi_score, life_expectancy, and education_index clearly indicate their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as train_model(), predict_hdi(), and load_dataset() are descriptive.
5	Code Comments	Inline and block comments explain logic	Yes	Comments explain preprocessing, model training, and prediction logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code is divided into reusable modules for preprocessing, training, prediction, and visualization.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed by using reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented using Python try-except blocks

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Model Training & Prediction	High

2	HDI Prediction Model	Scikit-learn	Prediction Service	High
3	Visualization Module	Matplotlib / Plotly	Dashboard & Reports	High
4	Database Module	SQLite / MySQL Medium	User Data & Prediction History	Medium
5	Authentication Module	Flask	Login & Session Management	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5/5	Well-organized project structure with separate modules.
Readability	5/5	Clear naming conventions and consistent formatting.
Reusability	5/5	Functions and modules are reusable across different features.
Documentation / Comments	4/5	Adequate comments with scope for more API documentation.
Overall Score	4.8/5	Clean, maintainable, and scalable code suitable for future enhancements.